Medicinal chemistry of NSAIDs

By

Gemmechu H (MSc)

Nonsteroidal anti-inflammatory drugs (NSAIDs)

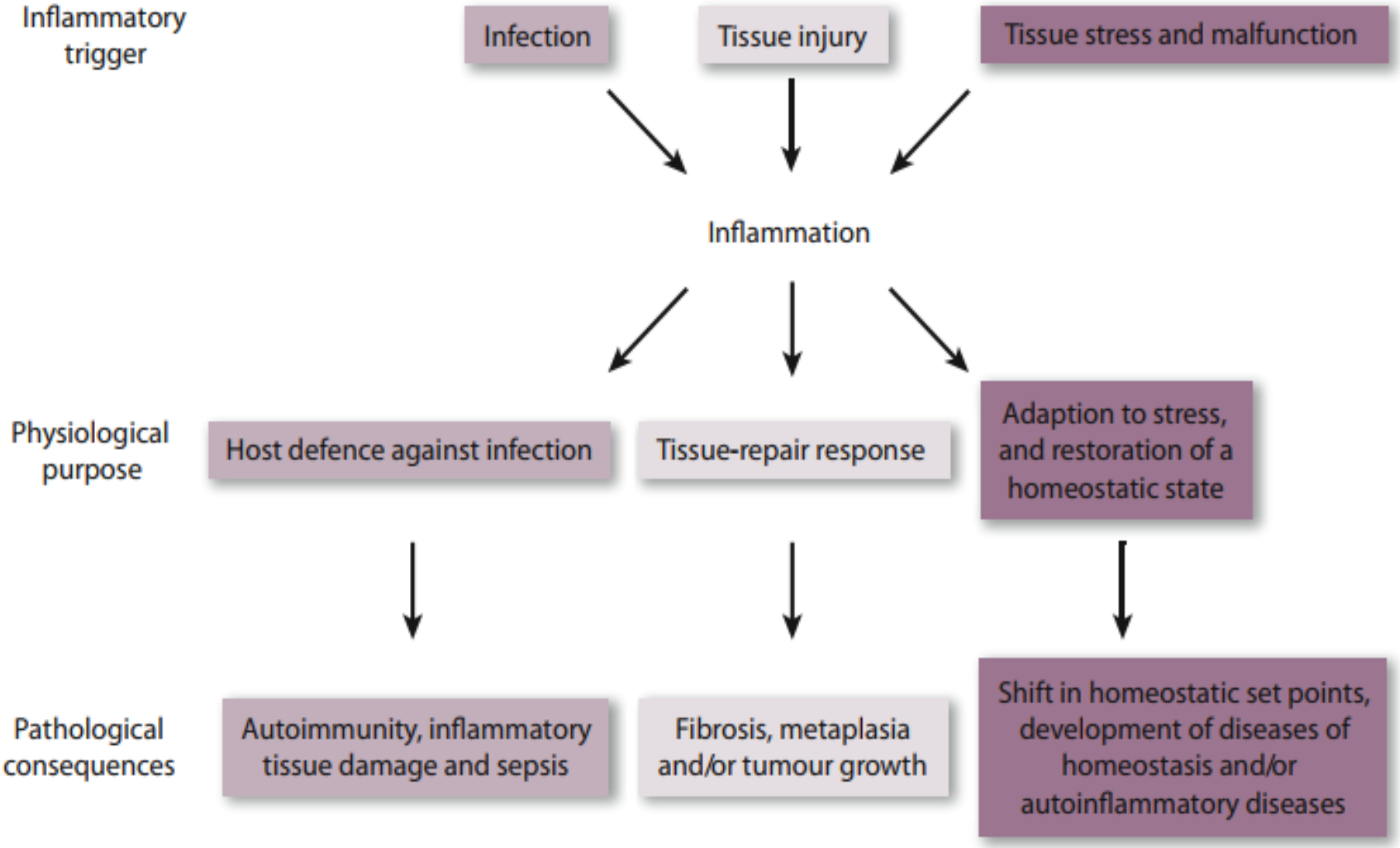
- NSAIDs are heterogeneous group of drugs having
  - Analgesic,
  - Anti-inflammatory and
  - Antipyretic effect.
- They represent one of the most widely used prescription and over-the-counter (OTC) drugs.
- Primarily used to treat inflammation, mild-to-moderate pain, and fever.
  - the treatment of headache, **arthritis**, sports injuries, and menstrual cramps and many other rheumatic diseases.

Cont'd…

- Produce anti-inflammatory action means they acts against inflammation.

  What is inflammation?

  - A normal and essential response to any noxious stimulus that threatens the host.



- Mediators of Inflammation: Eicosanoids

Mediators of Inflammation: Eicosanoids

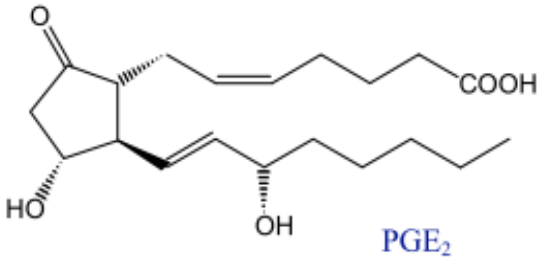
Prostaglandins and related compounds are collectively known as eicosanoids. They are not preformed but generated in-situ from phospholipids

Most are produced from **arachidonic acid**, a 20-carbon polyunsaturated fatty acid (5,8,11,14-eicosatetraenoic acid).

Examples of eicosanoids: prostaglandins, prostacyclins, thromboxanes, leukotrienes

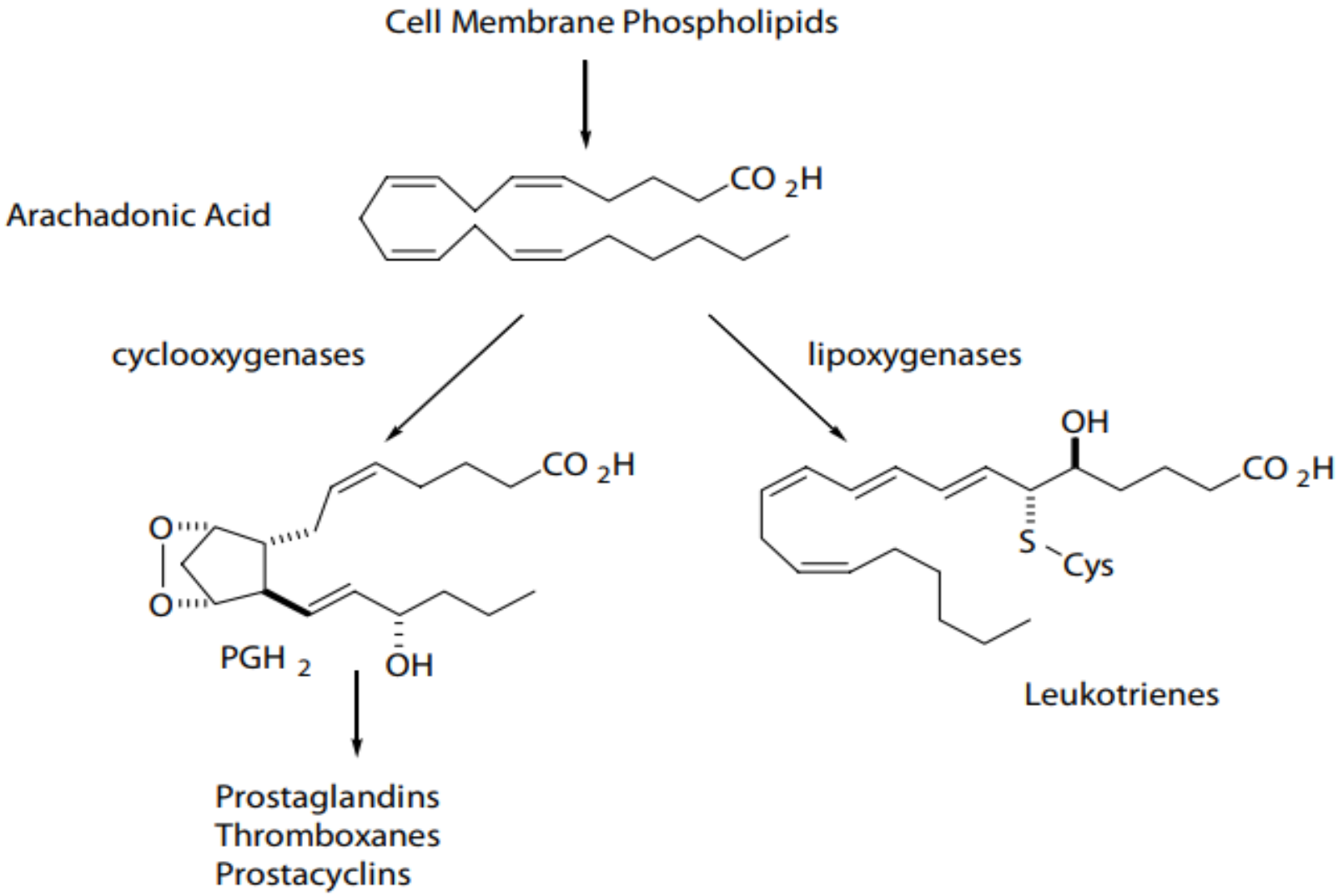
Prostaglandins all have a cyclopentane ring.

- A letter code is based on ring modifications (e.g., hydroxyl or keto groups).
- A subscript refers to the number of double bonds in the two sidechains.

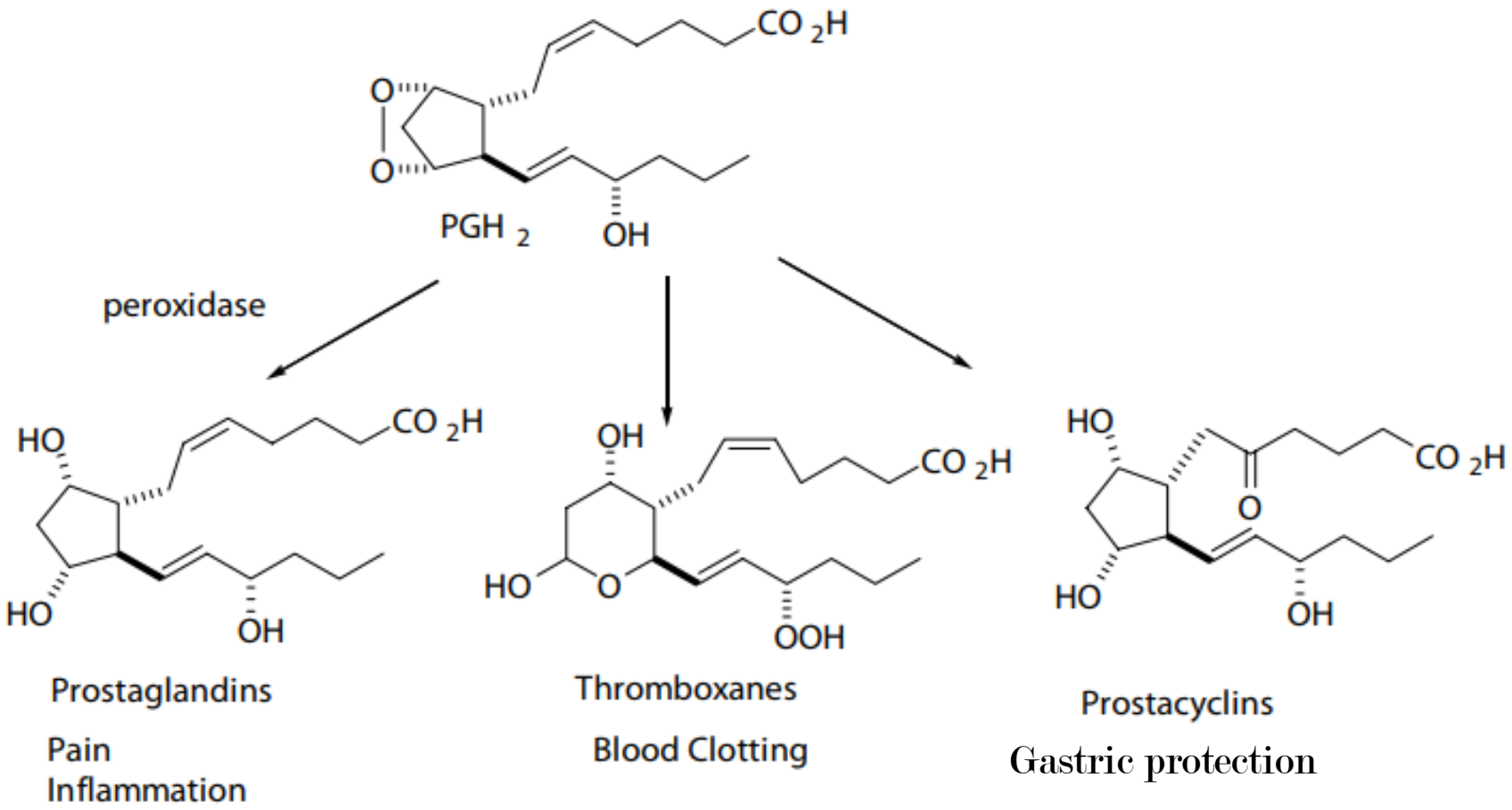


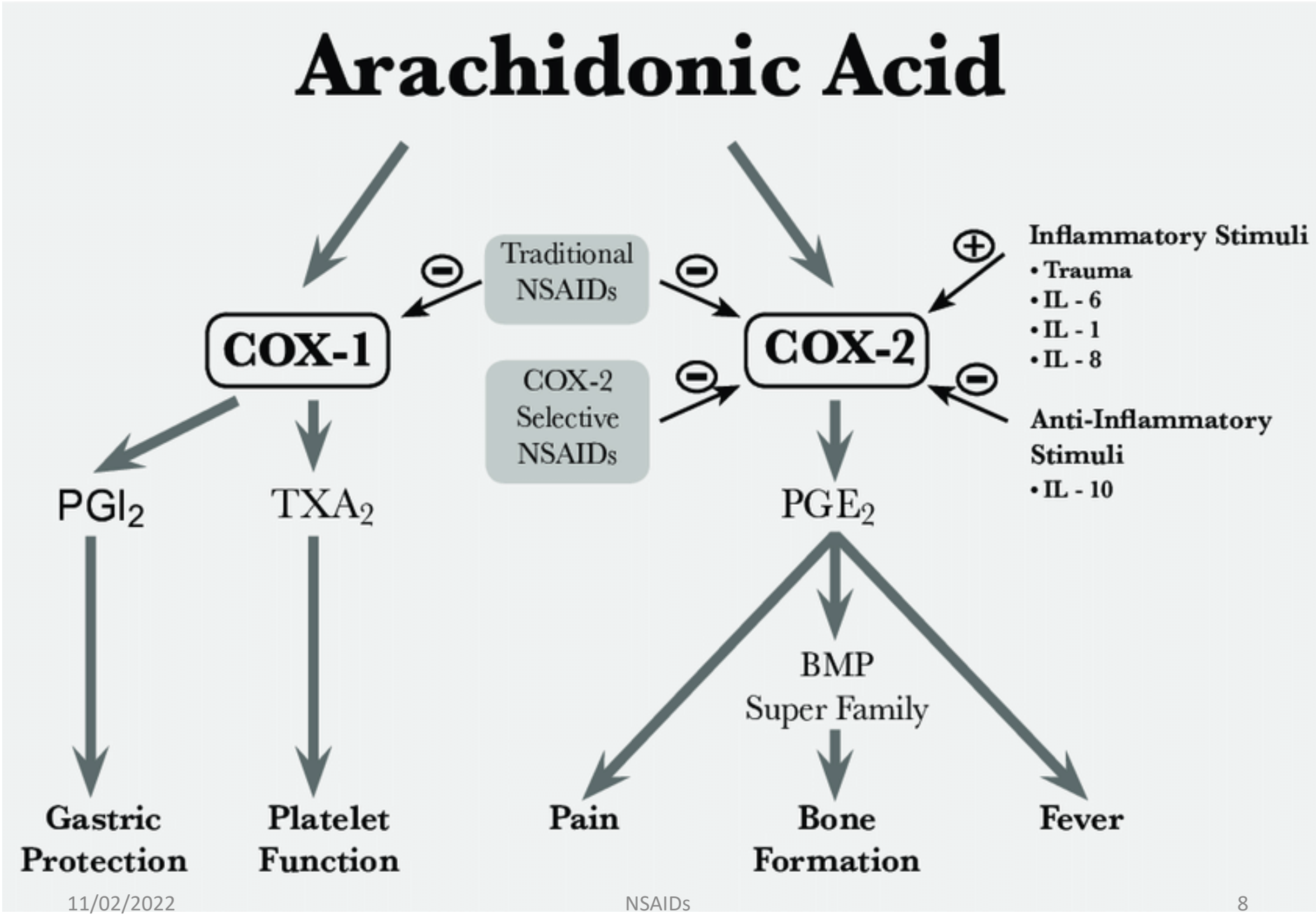
PGE_2

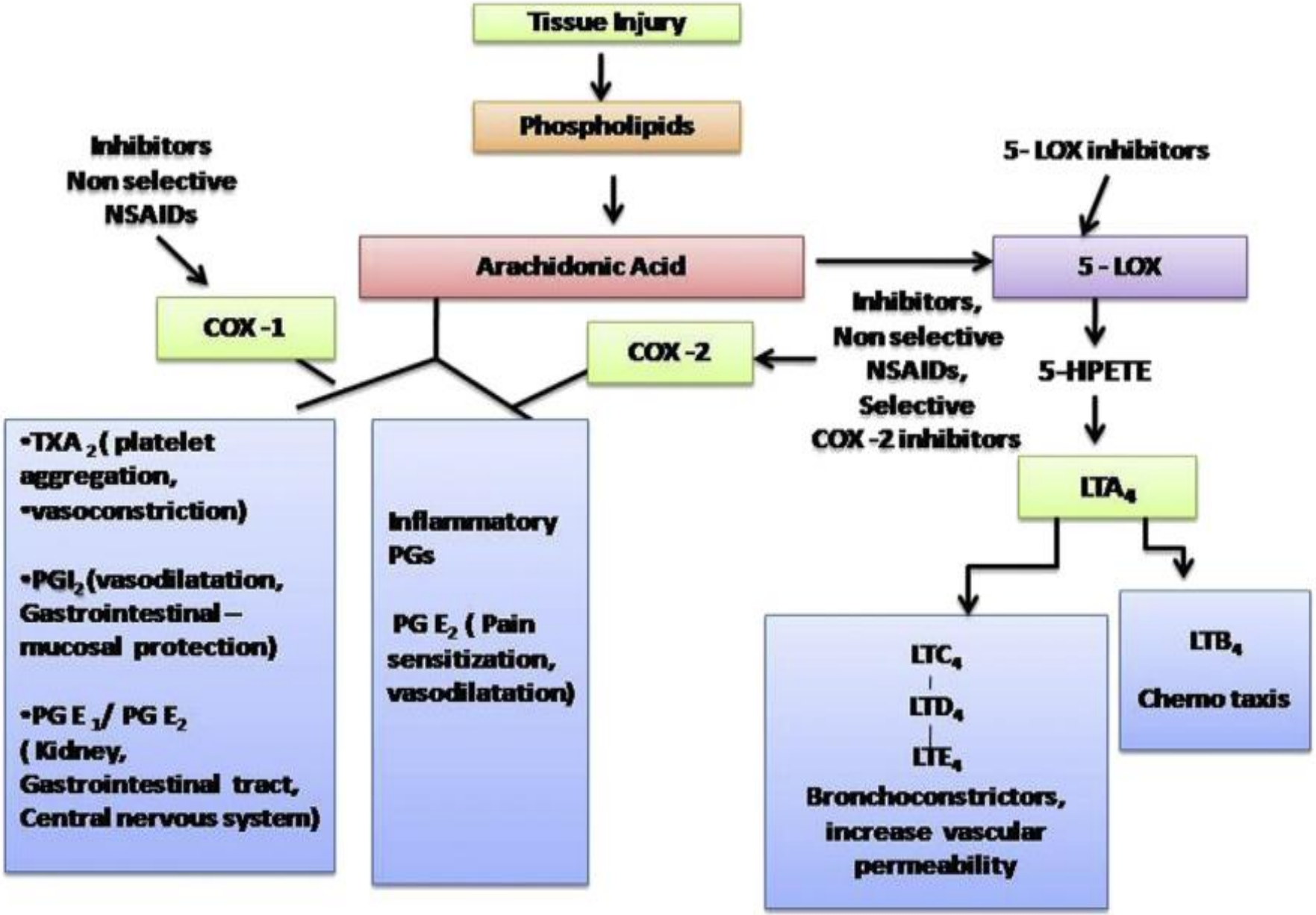
Thromboxanes are similar but have instead a 6-member ring.



- COX-1 is essential for thromboxane formation in blood platelets, and for maintaining integrity of the gastrointestinal epithelium.
- COX-2 levels increase in inflammatory diseases such as arthritis.


CO 2H
O
O
PGH 2
OH
peroxidase
HO
CO 2H
HO
OH
Prostaglandins
Pain
Inflammation
OH
CO 2H
HO
O
OOH
Thromboxanes
Blood Clotting
HO
CO 2H
O
HO
OH
Prostacyclins
Gastric protection


Arachidonic Acid
Traditional NSAIDs
COX-1
COX-2
COX-2 Selective NSAIDs
Inflammatory Stimuli
• Trauma
• IL - 6
• IL - 1
• IL - 8
Anti-Inflammatory Stimuli
• IL - 10
PGI2
TXA2
PGE2
BMP Super Family
Gastric Protection
Platelet Function
Pain
Bone Formation
Fever



COX – Cyclooxygenase; LOX – Lipooxygenase; LT – Leukotriene; PG s- Prostaglandins; PGI_2 – Prostacyclin ; TXA_2 – Thromboxane A_2; 5- HPETE – 5 hydroperoxyeicosatetraenoic acid

Side Effects

- In stomach
  - PGE2 and PGI2, serves as cytoprotective agents in gastric mucosa
- In platelets
  - formation of Thromboxane A2 (TXA2) in platelets
- In uterus
  - PGF2 is a potent uterotropic agent
- In kidney
  - decrease renal blood flow

Classification

- **Salicylic acid derivatives**
  - Aspirin, Diflunisal, Salsalate, Sulphasalazine
- **p-Amino phenol derivatives**
  - Paracetamol, Phenacetin
- **Pyrazolidinedione derivatives**
  - Phenyl butazone, Oxyphenbutazone, Sulphin-pyrazone.
- **Anthranilic acid derivatives**
  - Mefenemic acid, Flufenemic acid, Meclofenamate.

Classification…

- **Aryl alkanoic acid derivative**
  - Indole acetic acid derivatives
    - Indomethacin
  - Indeneacetic acid derivatives
    - Sulindac
  - Pyrrole acetic acid derivative
    - Tolmetin Sodium
  - Aryl and heteroaryl acetic/propionic acid derivatives
    - Ibuprofen, Ibufenac, Diclofenac, Naproxen, Fenoprofen, Ketoprofen, Flurbiprofen Caprofen, Ketorolac, Etodolac.

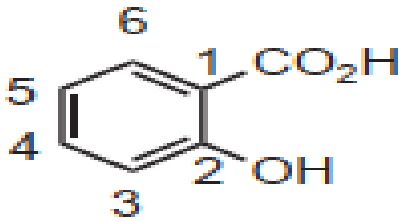
Classification…

- **Oxicams:** Piroxicam, Meloxicam
- **Selective COX-2 inhibitors:** Celecoxib, Rofecoxib, Valdecoxib.
- **Drug used in gout:** Allopurinoll, Probenecid, Sulphinpyrazone.

Salicylates

- Aspirin
- Salicylamide
- Salicylate Salts
- Salsalate
- Diflunisal

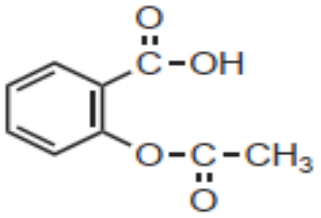
SAR of Salicylates



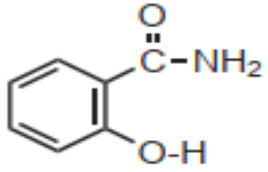
Salicylic acid

- Salicylate anion
- Reducing the acidity;
  - Analgesic actions Vs anti-inflammatory properties .
  - Eg-Salicylamide
- Substitution (carboxyl or phenolic).... potency and toxicity.
- Position of Phenolic hydroxyl to carboxyl group.
- Substitution of aromatic rings at the 5-position (e.g., diflunisal).

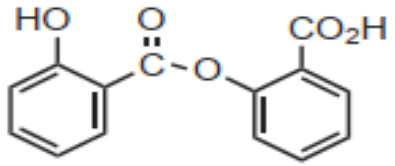
Specific drugs



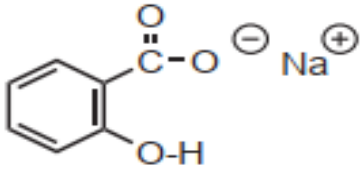
Acetylsalicylic acid



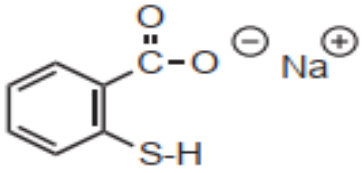
Salicylamide



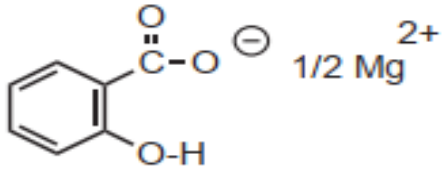
Salsalate (Disalcid)



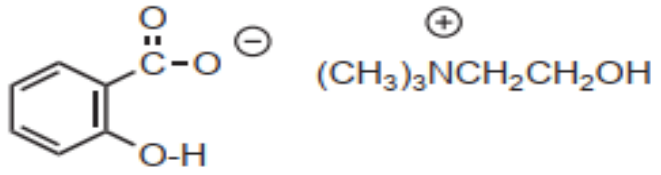
Sodium salicylate



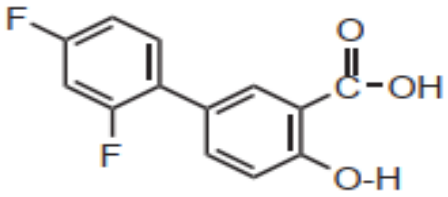
Sodium thiosalicylate



Magnesium salicylate



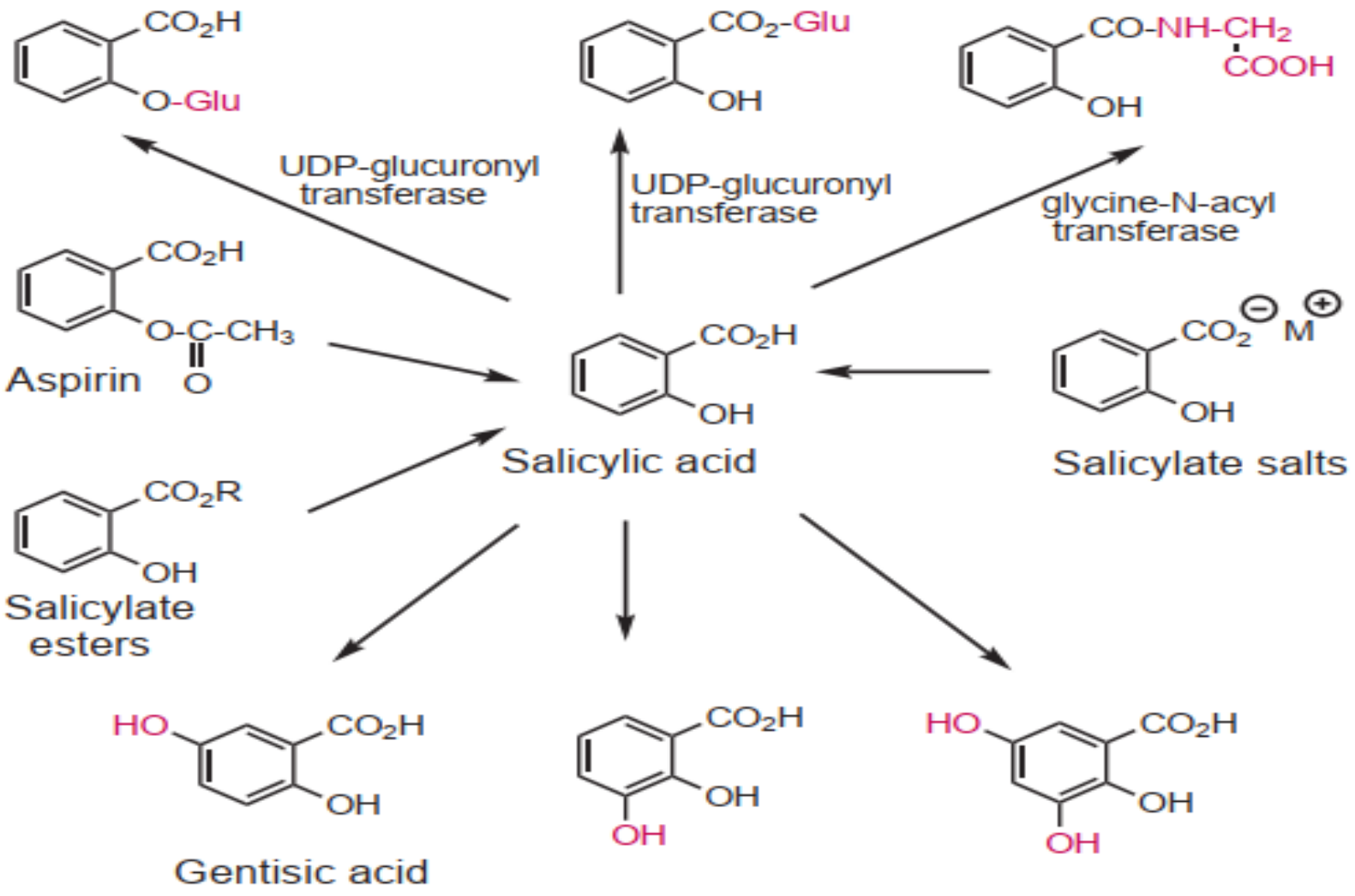
Choline salicylate



Diflunisal (Dolobid)

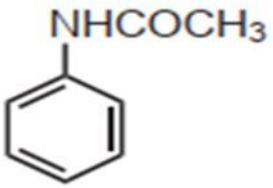
Figure: Structures of marketed derivatives of salicylic acid

Metabolism of Salicylate derivatives

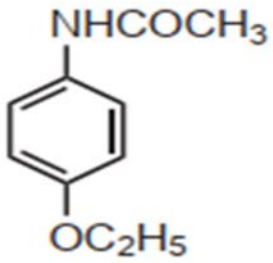


p-Amino phenol derivatives

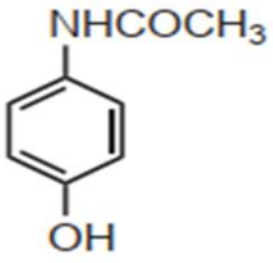
- Lack anti-inflammatory effects.
- Acetanilide and Phenacetin were introduced into the therapy in 1886 and in 1887 respectively.
  - too toxic
- Acetaminophen was introduced in 1893.



Acetanilide



Phenacetin

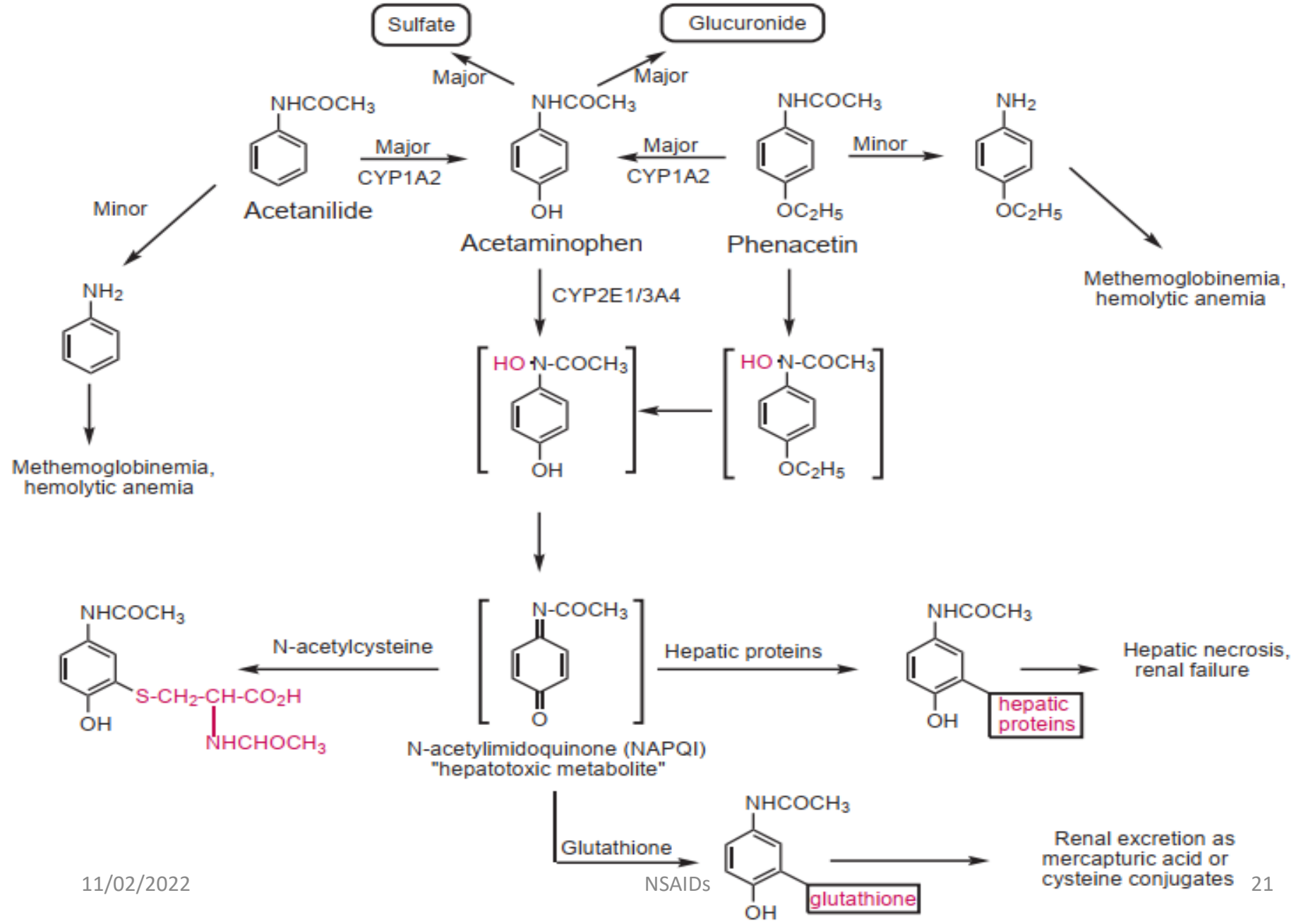


Acetaminophen

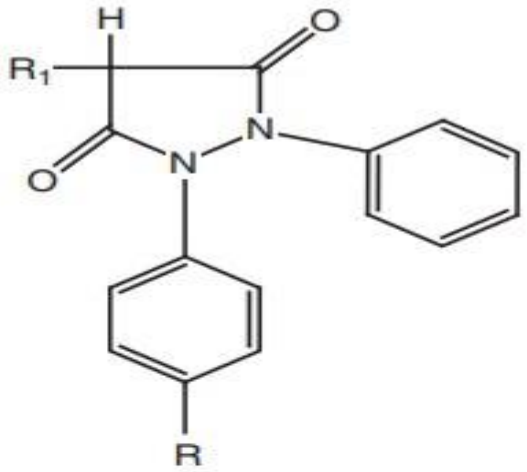
SAR of p-amino phenol derivatives

- Etherification of the phenolic function
  - methyl or propyl groups (side effects) vs ethyl groups .
- Substituents on the nitrogen atom that reduce basicity
  - metabolically labile (acetyl groups).

Metabolism of para aminophenol derivatives



3, 5-Pyrazolidinedione derivatives



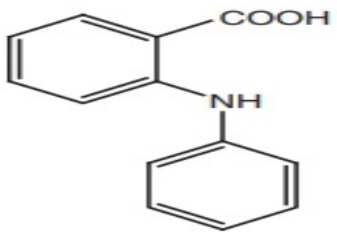
Name	R	R_1
Phenyl butazone	–H	$-C_4H_9$
Oxyphenbutazone	–OH	$-C_4H_9$
Sulphin pyrazone	–H	$-(CH_2)_2SOC_6H_5$

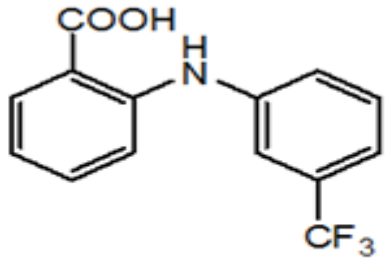
SAR of 3, 5-Pyrazolidinediones

- Replacement of one of the nitrogen atom in the pyrazolidinediones with an oxygen atom yields isoxazole analogues.
- In 3, 5-pyrazolidinedione derivatives, pharmacological activities are closely related to their acidity.
- Decreasing or eliminating acidity by removing the acidic proton at 4th position.
- Substitution of 2-phenyl thio ethyl group at the 4th position produces
  - antigout activity (sulphinpyrazone).

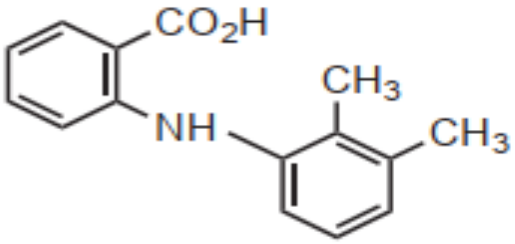
Anthranilic acid derivatives (Fenamates)

- Bioisosteric drug design concepts.

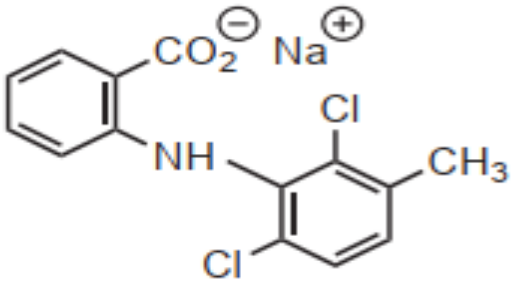




Flufenamic Acid ,

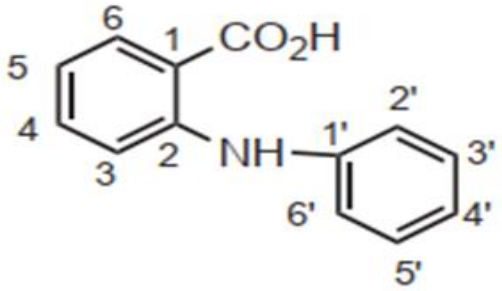


Mefenamic acid (Ponstel)



Meclofenamate sodium

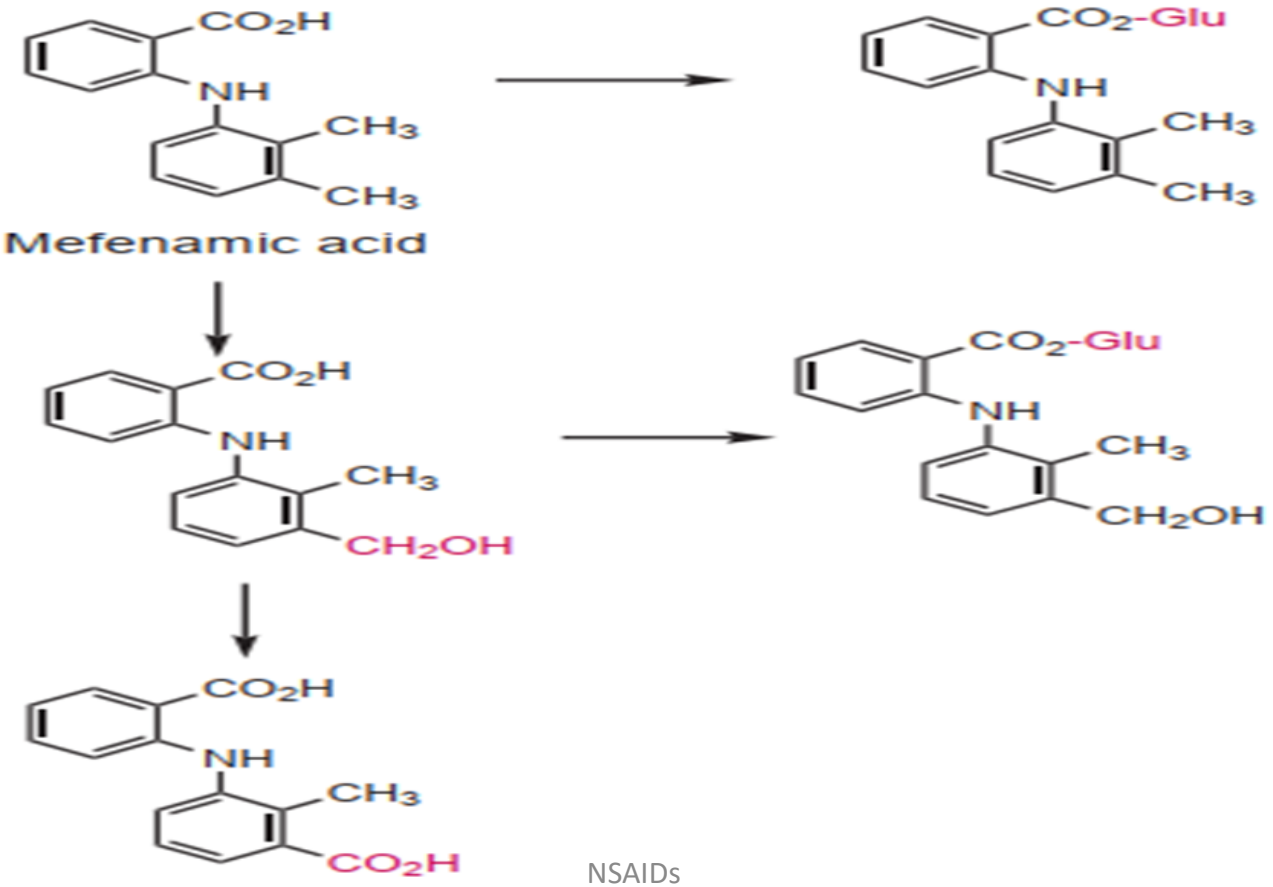
SAR of Anthranilic Acid Derivatives



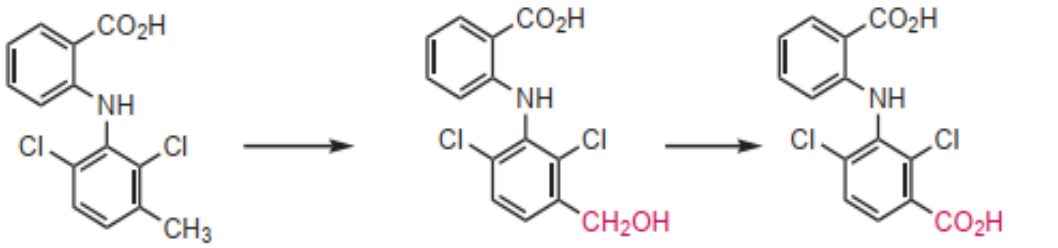
- The position of the carboxyl function to amino benzoic acid
  - (1,3 and 1,4)
- Substitution on the anthranilic acid ring Vs <u>N-aryl ring</u>.
  - 3′ › 2′ ›› 4′ for mono substitution with CF3 group (flufenamic acid).
  - 2′-Cl derivative being more potent than the 3′-Cl analog.
  - 2′,3′-disubstitution appears to be the most effective (mefenemic acid).
- The NH moiety of anthranilic acid is essential,
  - The replacement of NH function (O, CH2, S, SO2, N-CH3, or NCOCH3).
- Replacement of carboxylic acid function with the isosteric tetrazole (anti- inflammatory activity).

Metabolism of mefenamic acid

- Oxidation of the 3′-methyl group and glucuronidation of mefenamic acid and its metabolite .

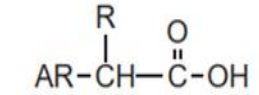


Metabolism of Meclofenamate Sodium



Meclofenamic acid 50-60% (active) 3-5% (inactive)

Arylalkanoic acids



R = H, CH_3 or alkyl
AR = aryl or heteroaryl

General SAR of Arylalkanoic Acids

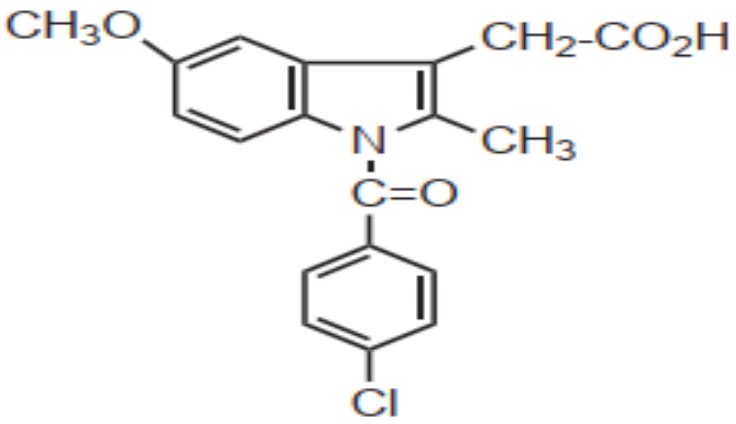
- All nonselective COX inhibitors possess a centre of acidity.
- The centre of acidity (one carbon atom adjacent to a flat surface).
- The distance between these centres is critical.
- Substitution of a methyl group on the carbon atom separating the acid center from aromatic ring (anti-inflammatory activity).
  - a-methyl acetic acid, or 2-substituted propionic acid ("profens").
  - groups larger than methyl decrease activity, alicyclic ring system!!
- Introduction of a methyl group creates a center of chirality.
  - anti-inflammatory activity (S-(+)- enantiomer).

Arylalkanoic acids-classification

- **Aryl and Heteroarylacetic Acids**
  - **Indole acetic acid derivatives (**Indomethacin)
  - **Indeneacetic acid derivatives (**Sulindac)
  - **Pyrrole acetic acid derivative (**Tolmetin , Zomepirac)
  - **Pyranocarboxylic acids derivative** (Etodolac)
- **Aryl and heteroaryl propionic acid derivatives**
  - Ibuprofen, Ibufenac, Diclofenac, Naproxen, Fenoprofen, Ketoprofen, Ketoprofen, Flurbiprofen, Caprofen, Ketorolac, Etodolac.

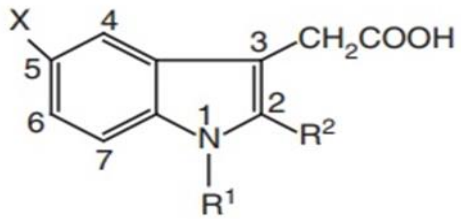
Indole acetic acid derivatives

- **Indomethacin**



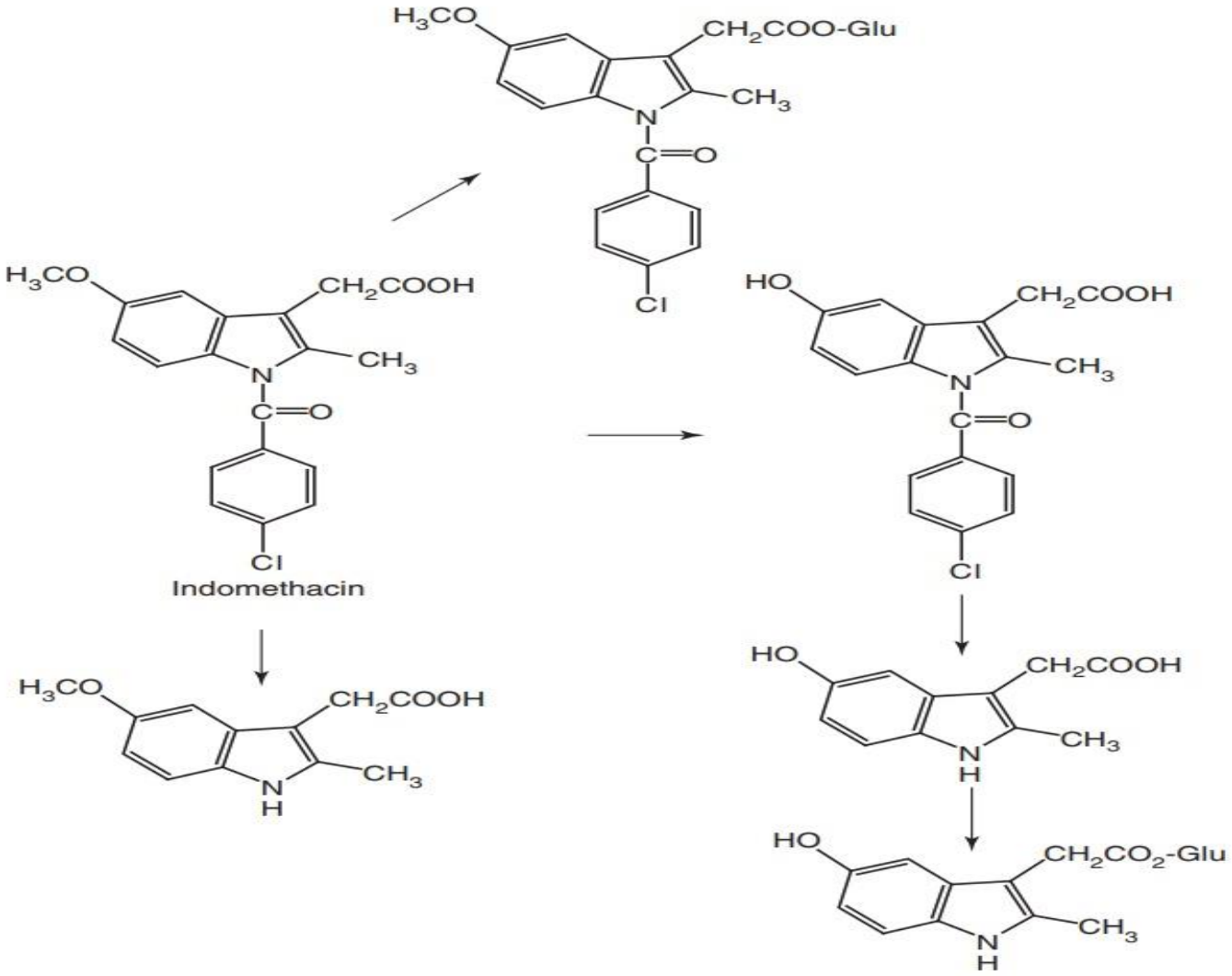
Indomethacin

SAR of Indole Acetic Acid Derivatives

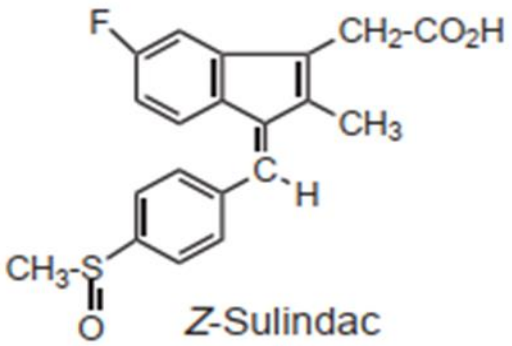


- Replacement of the carboxyl group with other acidic functionalities (the amide derivatives).
- Acylation of the indole nitrogen with aryl/alkyl carboxylic acids results in the decrease of activity.
- N-Benzoyl derivatives substituted with (fluoro, chloro, trifluoromethyl, or thiomethyl groups).
- Substituted indole ring Vs activity (methoxy, fluoro, dimethylamino, methyl, and acetyl).
  - X substituents activity are ranked as 5-OCH3 › N (CH3)2 › CH3 › H.
- The presence of indole ring nitrogen is not essential for activity (Sulindac).
- Substituents of R1 useful for increasing anti-inflammatory activity are ranked as C6H4CH2 › alkyl › H.
- Alkyl groups especially methyl group at α-position is much active than aryl substituted analogues.
- Substitution of a methyl group at the α position of the acetic acid side chain leads to equiactive analogues.
  - Chirality introduced in the molecules (S-(+)-enantiomer), anti-inflammatory activity.

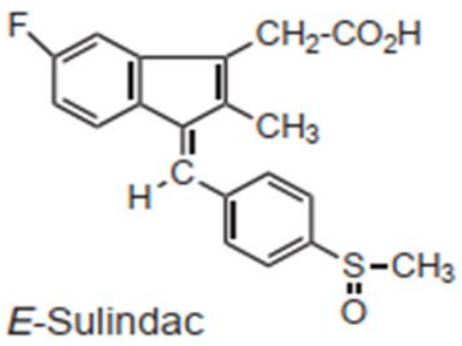
Metabolism of Indomethacin.



Indeneacetic acid derivatives



Z-Sulindac



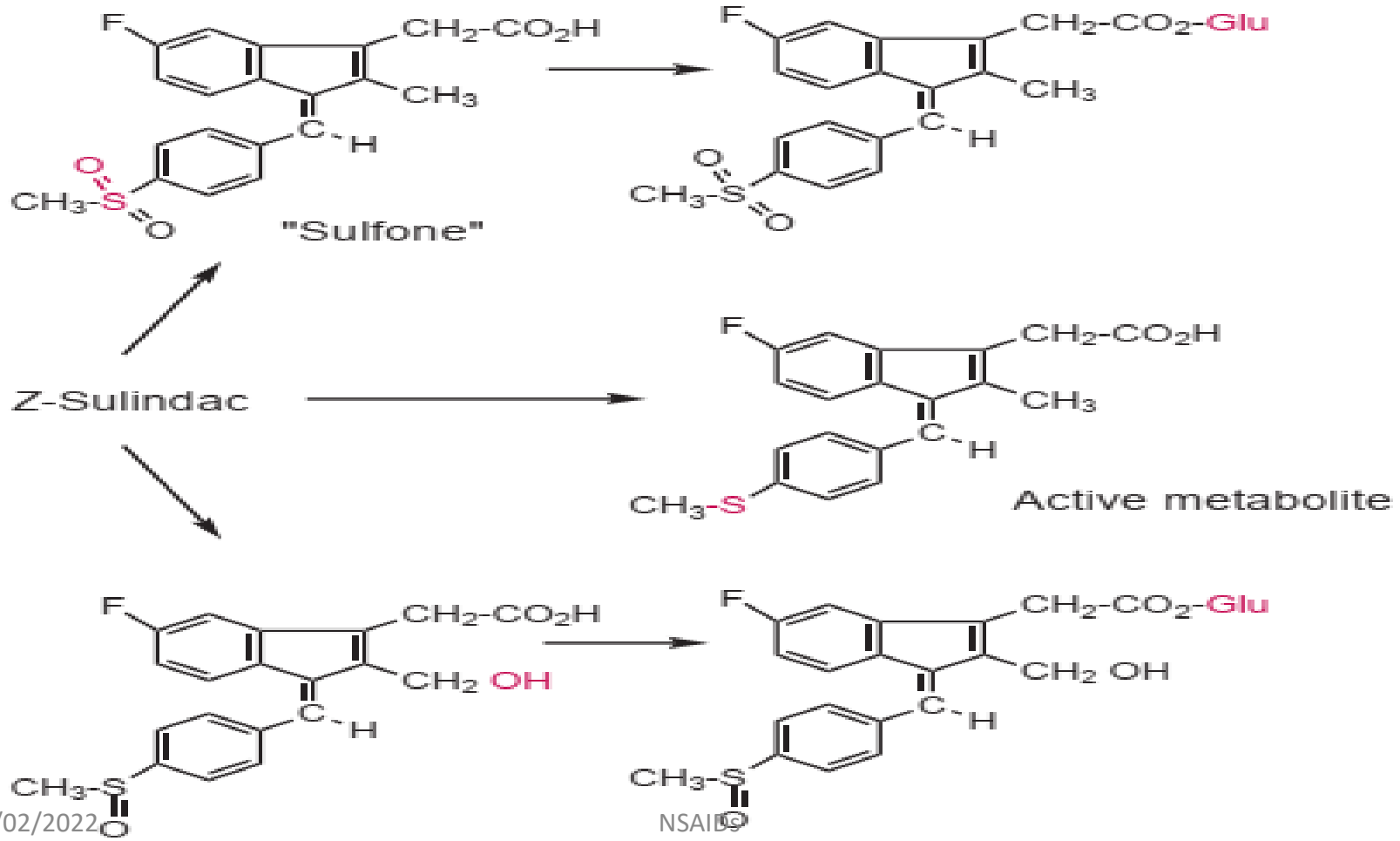
E-Sulindac

SAR of Indeneacetic acid derivatives

- The replacement of the indole ring with the indene ring system.
  - Therapeutically useful anti-inflammatory activity and fewer side effects but with other undesirable effects (poor water solubility).
- The replacement of the N-p-chlorobenzoyl substituent with a benzylidiene function (active).
- 5-methoxy group of the indene ring with a fluorine atom.
- Replacing the chlorine atom of the phenyl substituent with a sulfinyl group (water solubility).

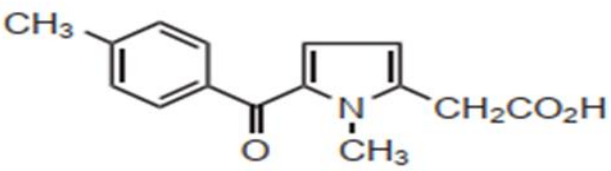
Metabolism of Sulindac

The major excretion product is the sulfone metabolite and its glucuronide conjugate.

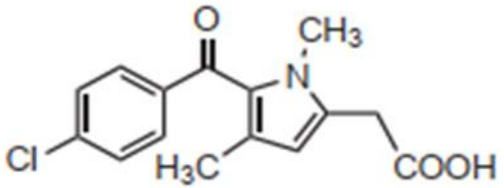


Pyrrole acetic acid derivative

- **Tolmetin, Zomepirac**



Tolmetin

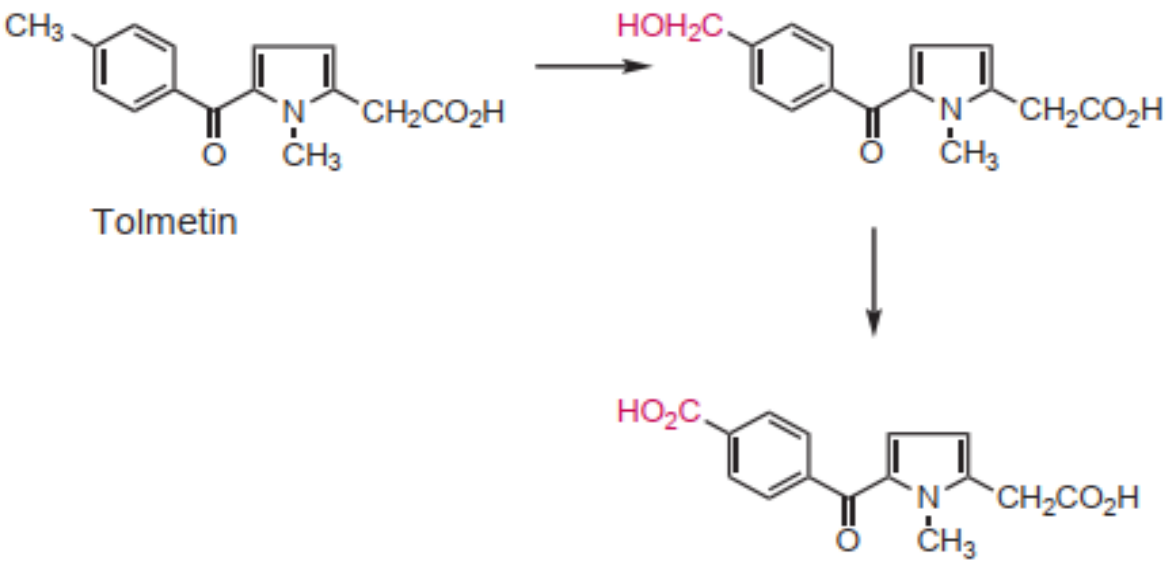


Zomepirac

SAR of Pyrrole Acetic Acid Derivative

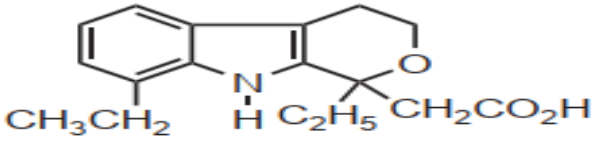
- Replacement of the 5-*p*-toluoyl group with a p-chlorobenzoyl moiety (little effect on activity).
- Methyl group in the 4-position of the pyrrole ring produced interesting results.
- Substitution of the p-methyl group of tolmetin with a p-chloro group blocked oxidative metabolism (Zomepirac).
    - Four times as potent as tolmetin (Withdrawn from the market)

Metabolism of tolmetin

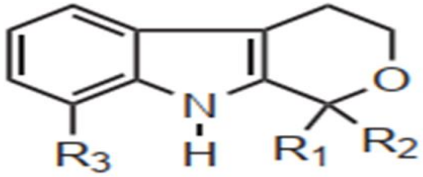


Pyranocarboxylic acids derivative

Etodolac

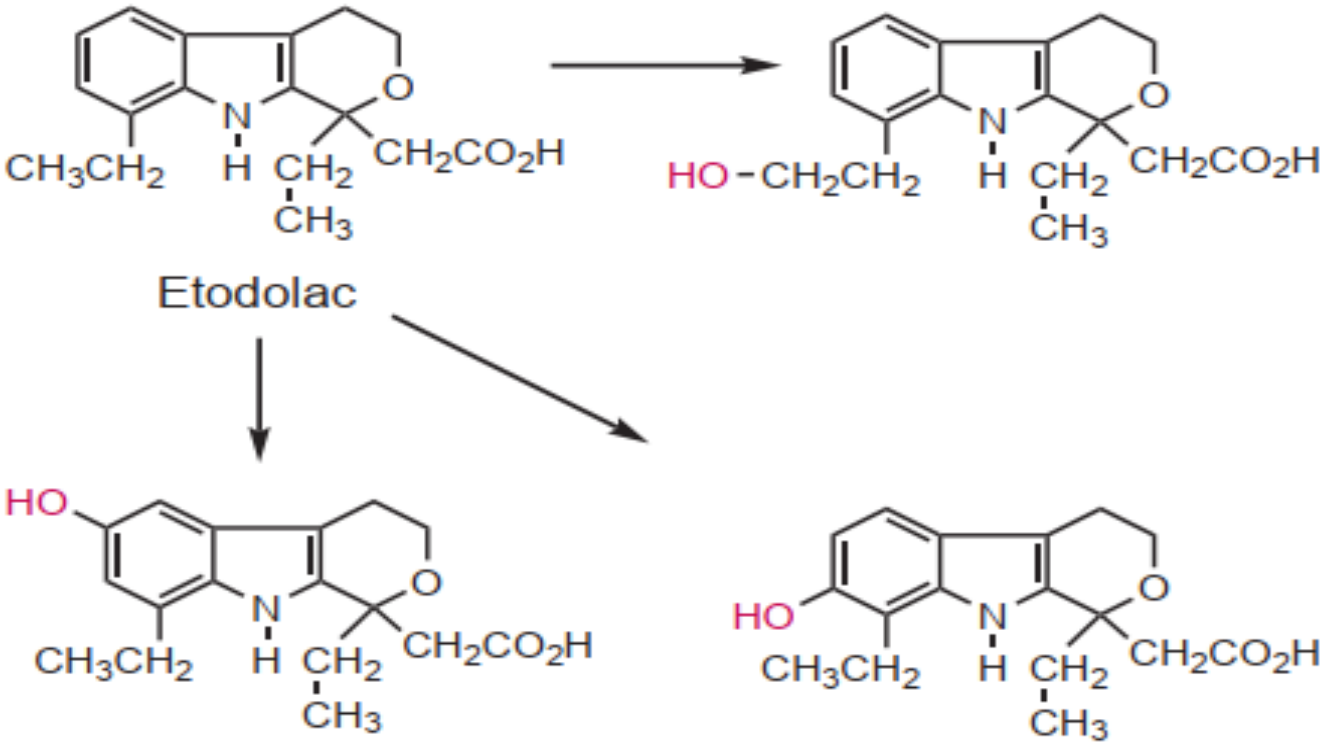


Etodolac

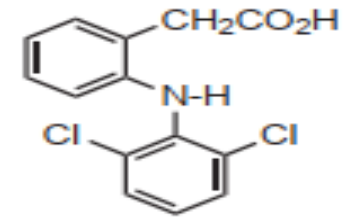


- Alkyl groups at R1 and an acetic acid function at R2.
- Chain length of the R1 substituent (ethyl or n-propyl).
- Aromatic substituents (8-position).
  - (8-ethyl, 8-n-propyl, and 7-fl uoro-8-methyl derivatives)

Metabolism of Etodolac



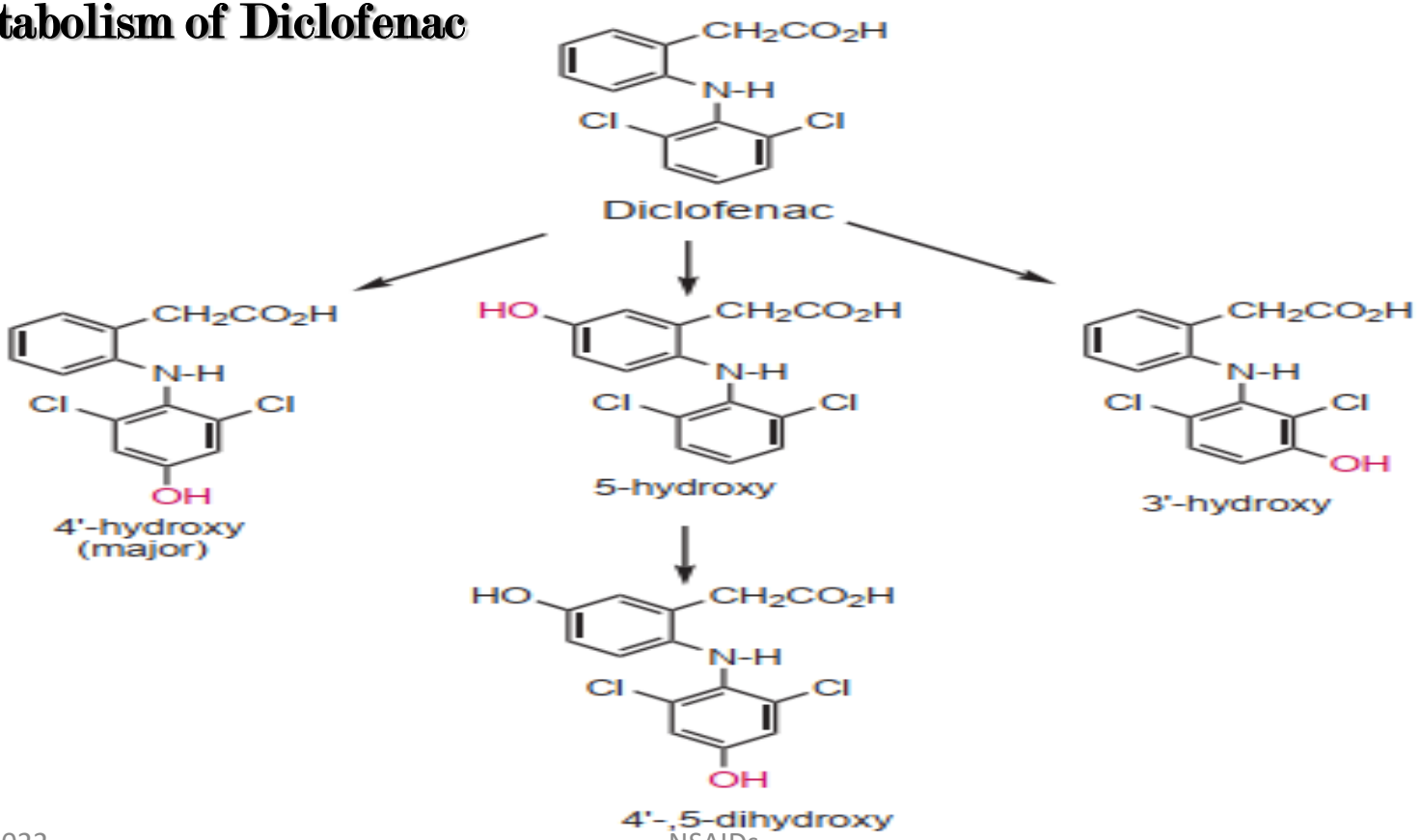
Diclofenac



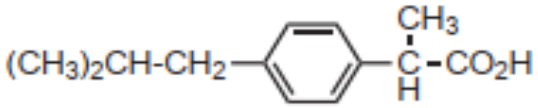
Diclofenac

- Both arylalkanoic acid and the anthranilic acid

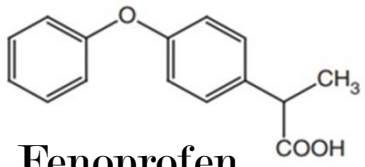
Metabolism of Diclofenac



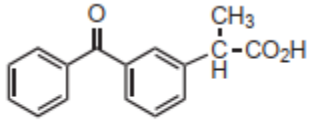
Aryl and heteroaryl propionic acid derivatives



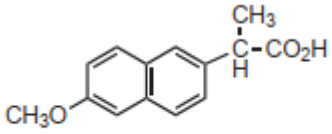
Ibuprofen



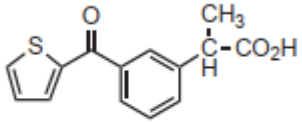
Fenoprofen



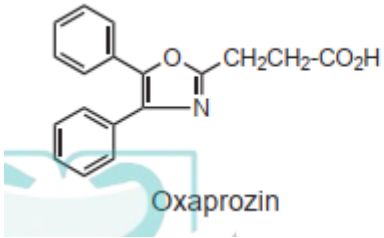
Ketoprofen



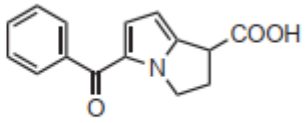
Naproxen



Suprofen



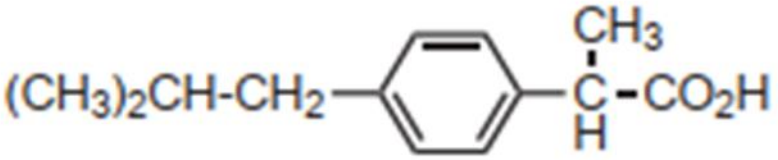
Oxaprozin



Ketorolac

- The existence of enantiomers, marketed as racemates

Ibuprofen

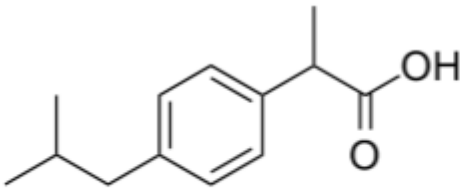


Ibuprofen

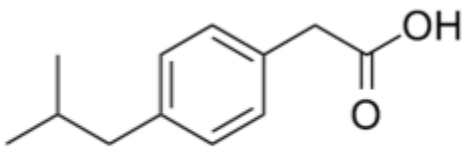
- OTC analgesic
- Marketed as the racemic mixture

SAR of Ibuprofen

- an *a*-methyl group on the alkanoicacid acid portion of acetic acid derivatives.
  - anti-inflammatory actions
  - side effects
  - The acetic acid analog of ibuprofen, ibufenac



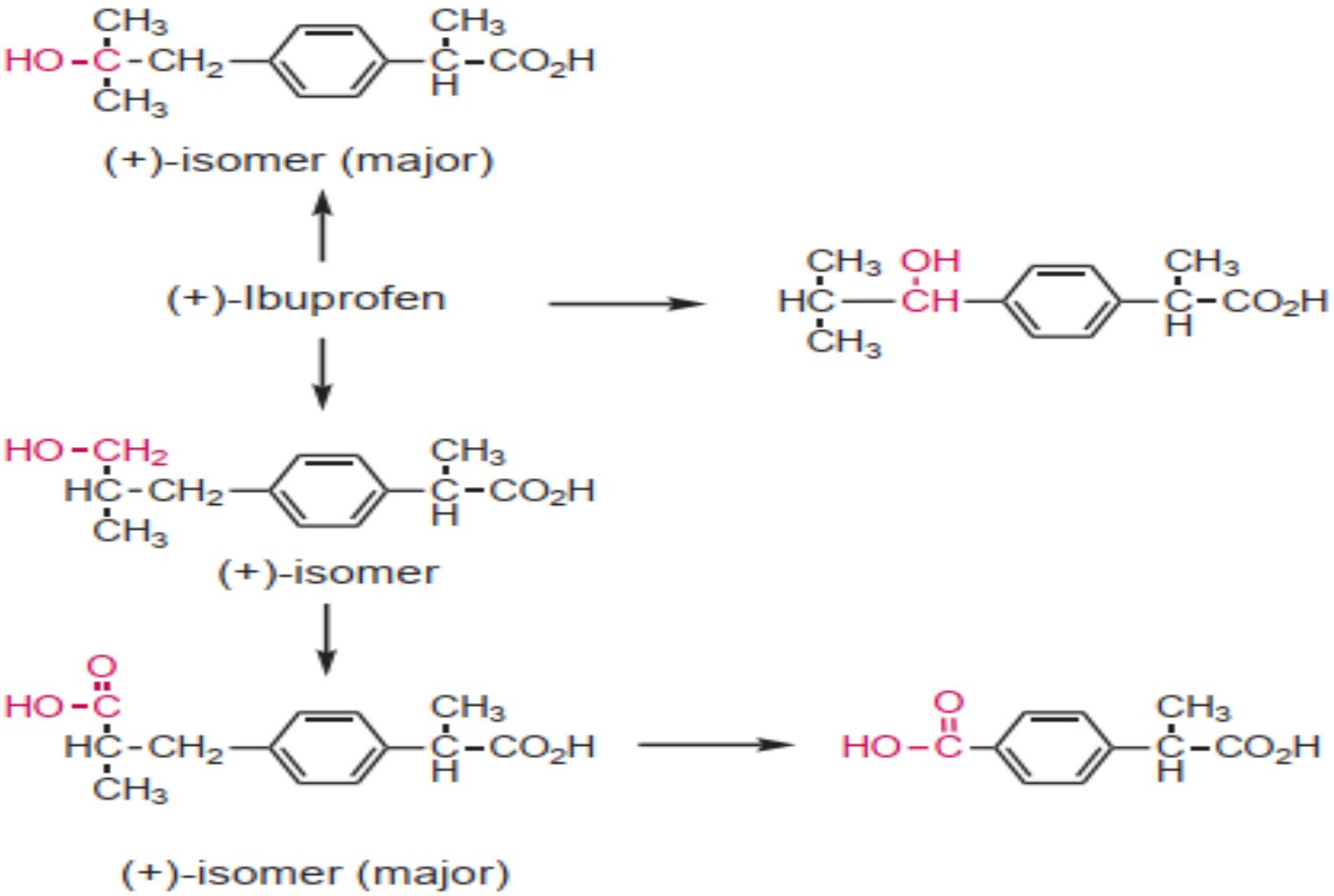
Ibuprofen

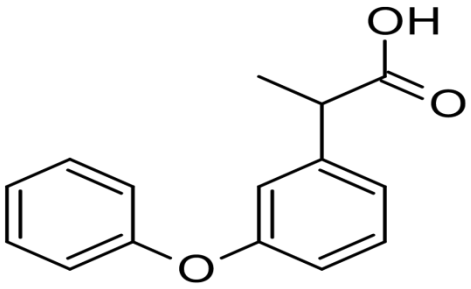


Ibufenac

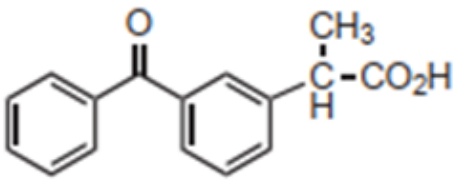
Metabolism of Ibuprofen

- Oxidation of the *p*-isobutyl side chain
  - alcohol oxidation to carboxylic acid (inactive)

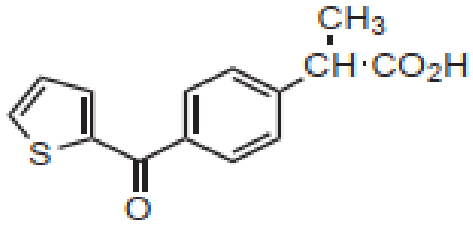




Fenoprofen



Ketoprofen

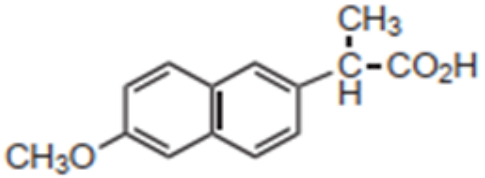


Suprofen

SAR of Fenoprofen

- Placing the phenoxy group in the ortho- or para-position of the arylpropionic acid ring .
- Replacement of the oxygen bridge between the two aromatic rings with a carbonyl group (ketoprofen).
  - thiophene isostere of ketoprofen
    - Suprofen (flank pain and transient renal failure)
- Extensively metabolized, primarily through glucuronide conjugation with the parent drug and
  - the 4′-hydroxy metabolite

SAR of Naproxen



Naproxen

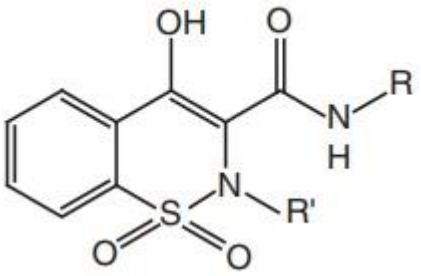
- In a series of substituted 2-naphthylacetic acids
  - substitution in the 6-position.
  - Small lipophilic groups
    - Cl, CH3S, and CHF2O, were active analogs (CH3O).
- Derivatives of 2-naphthylpropionic acids.
- Replacing the carboxyl group with functional groups capable of being metabolized to the carboxyl function (e.g., CO2CH3, — CHO, or — CH2OH).

Metabolism

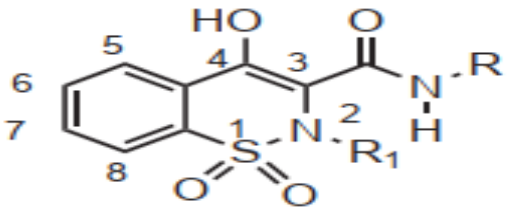
- Eliminated as either unchanged drug (60%) or as conjugates of unchanged drug (10%).
- The remainder is converted to the 6-O-desmethyl metabolite
  - further, to the glucuronide conjugate of the demethylated metabolite.

Oxicam

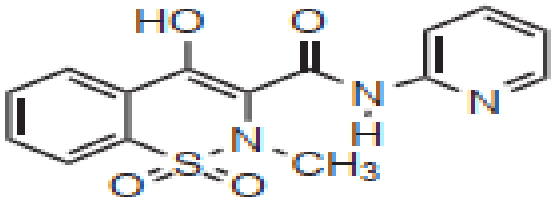
- The term oxicam described the relatively new enolic acid class of NSAIDs
  - 4-hydroxyl -1,2 benzothiazine carboxamide.



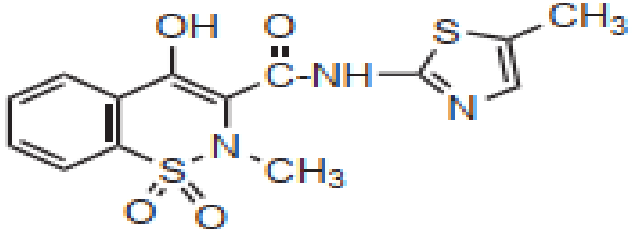
General Structure–Activity Relationships



- Methyl substituent on R1... optimum activity
- The carboxamide substituent, R
  - usually an aryl or heteroaryl substituent
  - alkyl substituents ???
  - The greatest activity (2-pyridyl, piroxicam; 2-thiazolyl, Meloxicam).

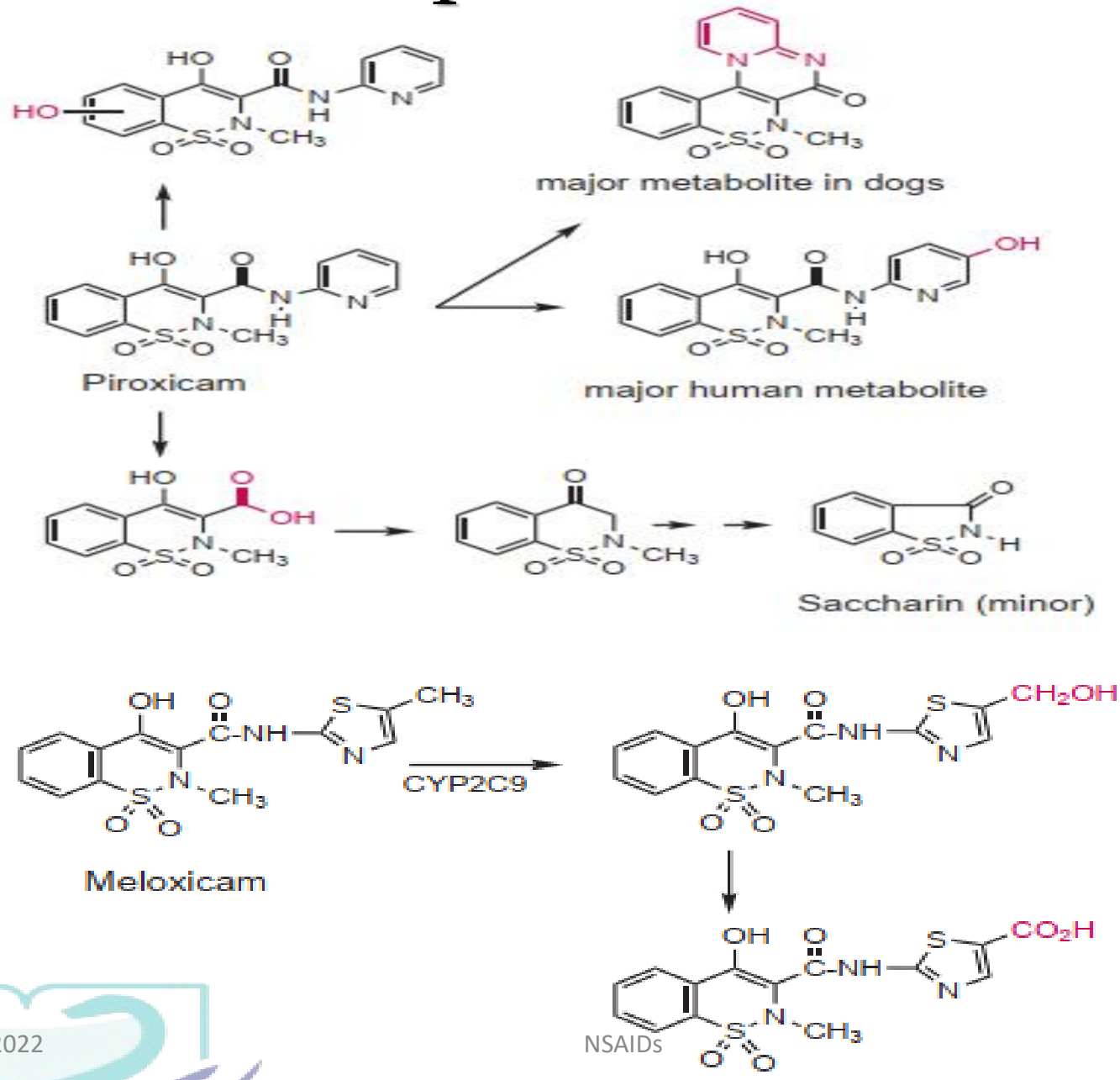


Piroxicam



Meloxicam

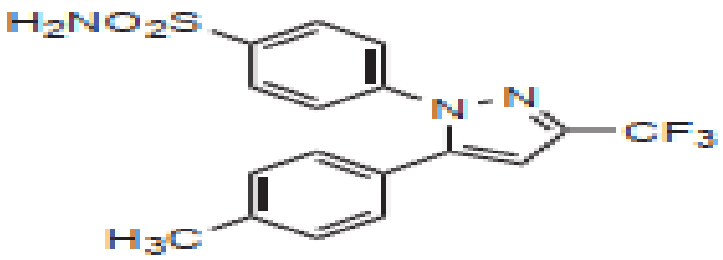
Metabolism of piroxicam and Meloxicam



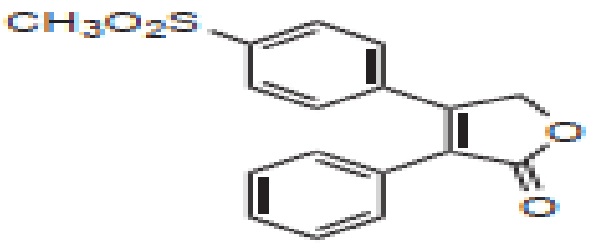
Selective COX-2 inhibitor

- Classic NSAIDs share similar side effect profiles,
  - particularly on the GI tract, many of which have been attributed to the inhibition of COX,
    - the rate-limiting step in prostaglandin biosynthesis.
- Extensive studies were conducted to search for selective COX-2 inhibitors.
  - (celecoxib, rofecoxib, lumiracoxib, and valdecoxib) are devoid of side effects, such as gastric ulcer.
  - but safety risks and cardiovascular side effects led to the removal of all coxibs except **celecoxib**.

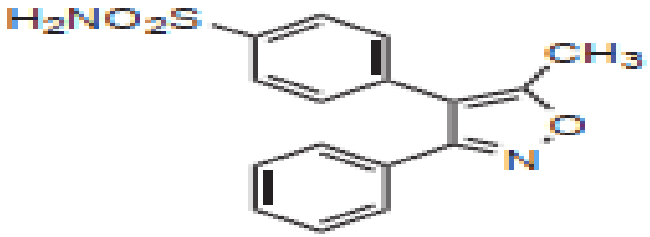
Specific Drugs



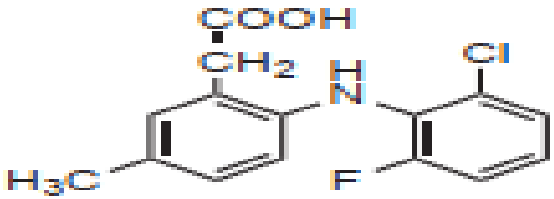
Celecoxib



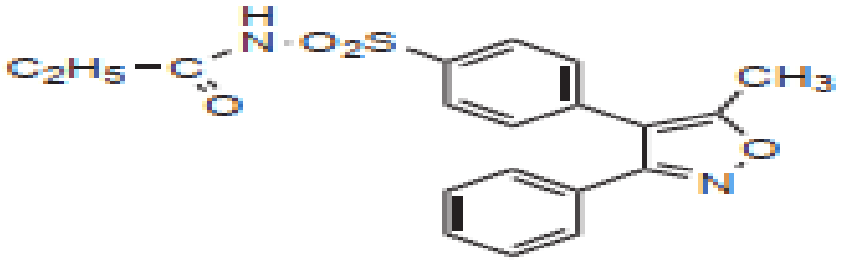
Rofecoxib



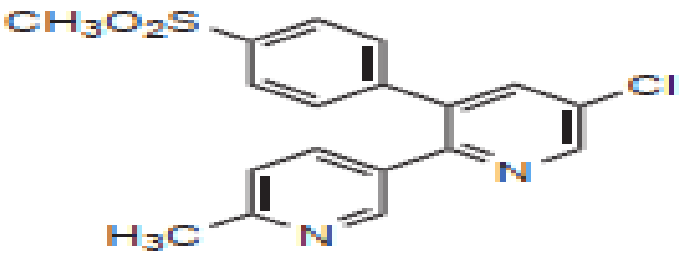
Valdecoxib



Lumiracoxib



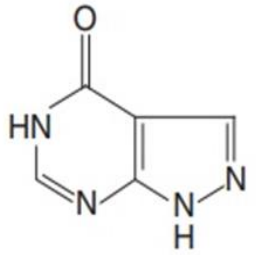
Parecoxib



Etoricoxib

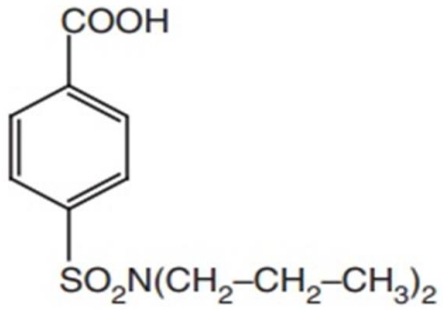
Drugs used in the treatment of gout

- An acute attack of gout occurs as a result of anti-inflammatory reaction to crystals of sodium ureate (the end product of purine metabolism in human beings) that is deposited in the joint tissues.
- Drugs used to treat gout may act in the following ways:
  - By inhibiting uric acid synthesis: Allopurinol.
  - By increasing uric acid excretion: Probenecid, Sulphinpyrazone.

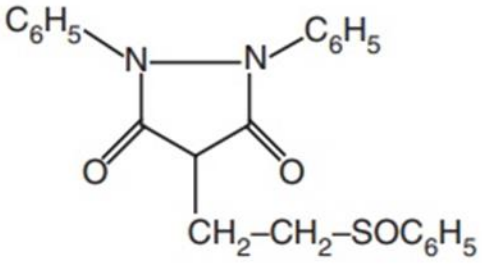


Pyrazolo pyrimidine-4-one

Allopurinol



Probenecid



Sulphinpyrazone

END!